



Laxmi Charitable Trust's
Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce
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Department of Information Technology (Bsc.IT Semester III)

Scala Programming Practical

Practical 19

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Practical 19

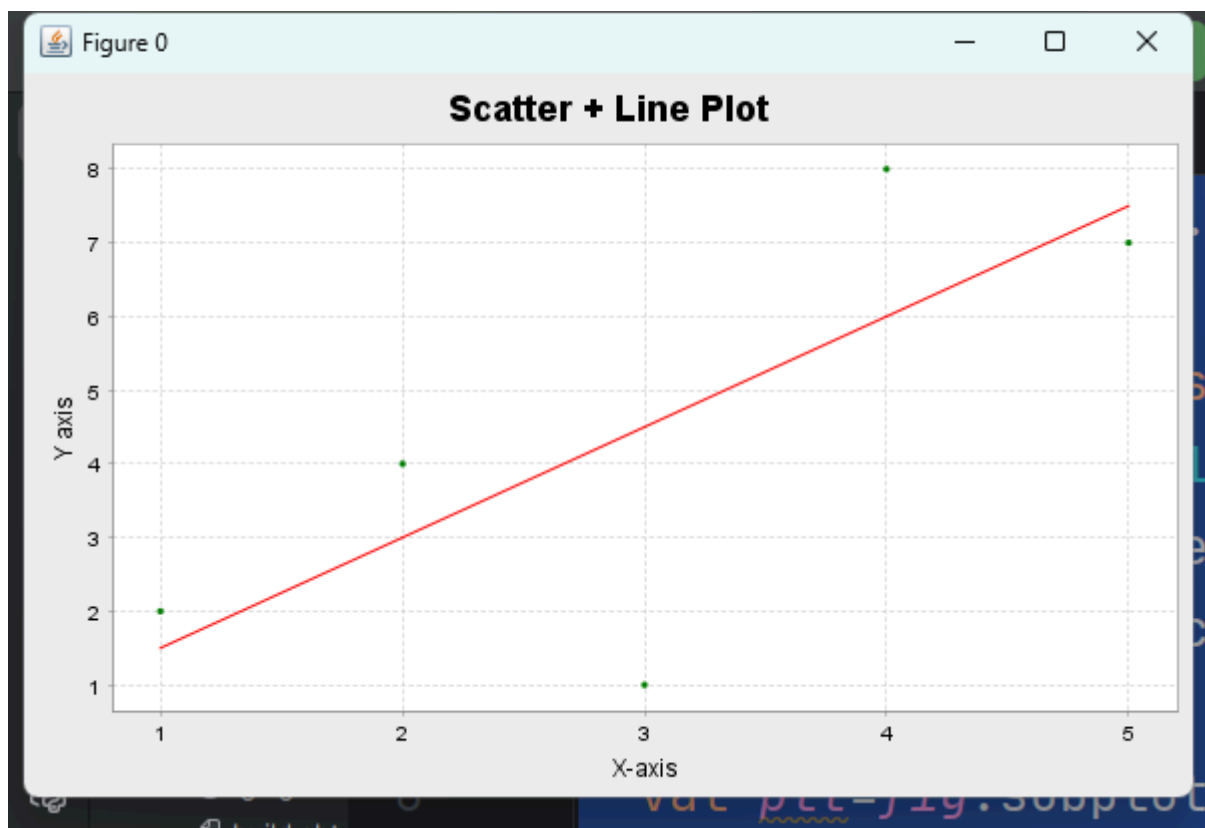
a)Combine Two Plots using Scatter and Line Plot in a single visualisation using breeze-viz

Code:

```
import breeze.linalg._
import breeze.plot._
object Prac14 extends App{
  val x=DenseVector(1.0,2.0,3.0,4.0,5.0)
  //Sample X-axis Values
  val yscat =
DenseVector(2.0,4.0,1.0,8.0,7.0)
  //Sample Y values for scatter
  val yline=
DenseVector(1.5,3.0,4.5,6.0,7.5)
  //Sample Y values for line plot
  val fig=Figure()
  val plt=fig.subplot(0)
```

```
plt += plot(x,  
yscat, '.',colorcode="Green")  
plt += plot(x,  
yline, '-',colorcode="red")  
plt.title="Scatter + Line Plot"  
plt.xlabel="X-axis"  
plt.ylabel = "Y axis"  
fig.refresh()  
println("Himanshu S037")  
}
```

Output:



Practical 20

Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.

Code:

```
import scala.io.Source

object Prac14 extends App {
  val file =
Source.fromResource("iris.csv")
  val data = file.getLines()
    .drop(1)
    .map(line =>
line.split(",")(0).toDouble)
    .toList

  file.close()

  val count = data.length
  val mean = data.sum / count
  val min = data.min
  val max = data.max

  val variance = data.map(x =>
math.pow(x - mean, 2)).sum / count
```

```
val std = math.sqrt(variance)

println(s"Count: $count")
println(f"Mean: $mean%.2f")
println(f"Minimum: $min%.2f")
println(f"Maximum: $max%.2f")
println(f"Standard Deviation:
$std%.2f")
println("Himanshu S037")
}
```

Output:

```
Count: 150
Mean: 75.50
Minimum: 1.00
Maximum: 150.00
Standard Deviation: 43.30
Himanshu S037
```